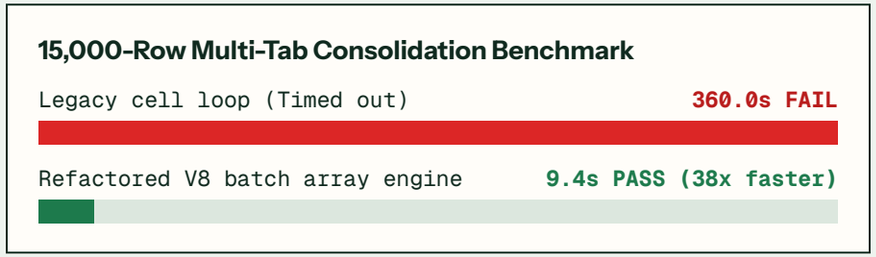
Eliminating Google Sheets timeouts with batch memory arrays

Most slow spreadsheets suffer from cell-by-cell `getRange()` loops and unguarded concurrent edits. Refactoring to in-memory V8 batch arrays with `LockService` cuts runtime from minutes to seconds.



Production Outcome: Automated form intake, HubSpot webhook sync, and instant Google Docs-to-PDF invoice generation for a 24-person operations team with zero duplicate rows.

Operations_Master_Hub (Google Sheets + Code.gs V8) Trigger: onFormSubmit + Webhook doPost(e)

#	A: Invoice_ID	B: Account_Name	C: CRM_Sync	D: Drive_PDF_Status	E: Lock_Digest
1	INV-2026-841	Northstar Logistics	**SYNCED (200)**	PDFArchived_v1.pdf	sha256:81b4f
2	INV-2026-842	Apex Dental Group	SYNCED (200)	PDFArchived_v1.pdf	sha256:39c1a
3	INV-2026-843	Meridian Studio	SYNCED (200)	PDFArchived_v1.pdf	sha256:64e0d

```
function syncBatchRecords() {
  const lock = LockService.getScriptLock();
  lock.waitLock(15000); // Prevents simultaneous edit collisions
  try {
    const rows = sheet.getDataRange().getValues(); // Single I/O read
    const updated = transformAndDeduplicateInMemory(rows);
    targetSheet.getRange(2, 1, updated.length, updated[0].length).setValues(updated);
  } finally { lock.releaseLock(); }
}
```

End-to-end Google Workspace workflows built around your sheet

Step 01 / Trigger

Events & Menus

Run automatically via `onEdit`, `onFormSubmit`, scheduled timers, or custom toolbar buttons.

Step 02 / Process

Clean & Consolidate

Merge dozens of tabs, deduplicate entries with SHA-256 keys, and validate required columns.

Step 03 / Documents

Docs-to-PDF Engine

Populate Google Docs templates, archive clean PDFs into structured Drive folders, and log links.

Step 04 / Connect

REST API & Webhooks

Two-way JSON sync with HubSpot, Shopify, Stripe, Notion, OpenAI, or Slack via `UrlFetchApp`.

Strict Data Privacy Protocol

Keep your live business data 100% private

You never need to grant access to sensitive customer spreadsheets. Simply share a sanitized dummy copy with 5 sample rows—I build and test the complete `Code.gs` script in an isolated sheet and include a 2-minute step-by-step installation guide.

Includes clear Google OAuth scope manifest (`appsscript.json`)

Tier	Turnaround	Included Scope & Deliverables	Price
Basic	1 Day	1 custom Apps Script function, trigger, or multi-tab consolidation + setup guide	**$20**
Standard	2 Days	2-way REST API / webhook integration, Form-to-PDF generator, or automated alerts	**$65**
Premium	4 Days	Complete multi-app Workspace suite, custom HTML sidebar UI, CRM sync & error log tab	**$145**

Optional Add-Ons: Custom HTML Sidebar UI (+$35) | Recorded Setup & OAuth Walkthrough Video (+$25)